COMPUTER PROGRAMMING • MIDTERM

Prime Factorization

DESCRIPTION

Given an integer **n**, write it as a product of prime numbers. List the primes in increasing order. If a prime p occurs e times and e > 1, write it as `p^e`; if it occurs only once, write just `p` (never `p^1`). Separate the factors with `*` surrounded by one space on each side. If **n** is itself a prime, print **n**.

INPUT FORMAT

The input is a single line containing one integer **n** (1 < **n** ≤ 100000000). There is exactly one test case; do not read until end of file.

OUTPUT FORMAT

Print the factorization on one line, in the form shown in the samples, followed by a newline. Use `^` only for exponents above 1, and put exactly one space before and after every `*`.

SAMPLE

Sample Input	Sample Output
20	2^2 * 5
17	17
360	2^3 * 3^2 * 5